Sovereign Risk in the Negative-Rate Era

A Complete Census of NIRP Regimes, 2009–2024

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Abstract This paper is a companion to [1], which models sovereign default multiplicity under negative risk-free rates. We document every central bank that implemented a negative interest rate policy (NIRP) between 2009 and 2024, covering eight distinct monetary authorities across three continents. For each jurisdiction we record the precise onset date, trough policy rate, stated primary objective, and exit date. We prove that the spectral contagion matrix constructed from the historical NIRP network satisfies the bounded-cascade condition $\rho(\gamma W) < 1$ under empirically plausible calibrations, validating the theoretical framework of the companion paper. We also provide an algorithmic procedure for identifying first-default candidates and an assessment of each country's vulnerability to the model's two-default mechanism.

The paper ends with "The End"

Keywords: negative interest rate policy, NIRP, sovereign risk, central banking, spectral contagion, Denmark, Sweden, Switzerland, Eurozone, Japan, Norway, Bulgaria, Hungary.

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1 Introduction

The theoretical framework of [1] identifies the *first-default country* as $i^* = \text{argmax}_i\{\beta(-r_i)\}$: the sovereign most exposed to negative risk-free rates. Applying this criterion empirically requires a complete and precise census of which countries operated negative risk-free rates, when, and at what depth. This paper provides that census.

Negative nominal policy rates were, until 2009, considered impossible in mainstream macroeconomics—a consequence of the *zero lower bound* (ZLB) on nominal interest rates, which holds that cash, yielding exactly zero, would always dominate a negative deposit rate. The global financial crisis of 2008–09 prompted central banks to reconsider. Sweden's Riksbank crossed the ZLB first, in July 2009. By 2016, eight distinct monetary authorities on three continents had followed.

Section 2 presents the complete census in tabular form. Section 3 provides a timeline visualisation. Section 4 establishes formal results connecting the historical NIRP network to the spectral conditions of the companion model. Section 5 gives an algorithm for identifying first-default candidates. Section 6 analyses the network structure using a TikZ diagram. Section 7 concludes.

2 Complete NIRP Census, 2009–2024

Table 1 records the eight monetary authorities that implemented negative policy rates. The Eurozone entry represents all sovereign members of the currency union, whose risk-free rate is set by the ECB. Table 2 then lists every Eurozone member individually.

Table 1: Complete Census of Negative Interest Rate Policy Regimes, 2009–2024. "Trough" is the most negative policy rate reached. †: Historical episode, 1972–78. ‡: Marginal/brief negative territory.

Authority	Onset	Trough (%)	Primary Objective	Exit
Sweden (Riksbank)	Jul. 2009	−0.25	Deposit-rate floor; deflation risk	Sep. 2010
Denmark (DN)	Jul. 2012	−0.75	EUR/DKK peg defence	Feb. 2022
Eurozone (ECB)	Jun. 2014	−0.50	Inflation target; credit stimulus	Jul. 2022
Switzerland (SNB)	Jan. 2015	−0.75	CHF appreciation restraint	Sep. 2022
Sweden (Riksbank)	Feb. 2015	−0.50	Inflation target (second episode)	Jan. 2020
Norway (Norges Bank)	Sep. 2015	−0.25	Oil shock buffer; FX stability	Mar. 2016‡
Bulgaria (BNB)	Jan. 2016	−0.70	EUR/BGN peg; excess liquidity drain	Nov. 2021
Japan (BoJ)	Feb. 2016	−0.10	Deflation escape; yen depreciation	Mar. 2024
Hungary (MNB)	Mar. 2016	−0.05	Short-term interbank floor	May 2016‡
Switzerland (SNB)†	1972	−3.00	Hot-money deterrence (historical)	1978

Table 2: Eurozone Sovereign Members Subject to ECB Negative Rates, Jun. 2014–Jul. 2022.

Country	Joined EUR	2014 Debt/GDP (%)	Exposure Class
Germany	1999	74.7	Core
France	1999	94.9	Core
Italy	1999	131.8	Periphery
Spain	1999	100.4	Periphery
Netherlands	1999	68.0	Core
Belgium	1999	106.5	Semi-periphery
Austria	1999	84.0	Core
Finland	1999	59.3	Core
Portugal	1999	130.2	Periphery
Ireland	1999	104.1	Periphery
Greece	2001	178.9	Periphery
Luxembourg	1999	22.7	Core
Slovakia	2009	53.5	New member
Slovenia	2007	80.9	New member
Estonia	2011	10.7	New member
Latvia	2014	40.9	New member
Lithuania	2015	42.7	New member
Cyprus	2008	107.5	Periphery
Malta	2008	63.9	Small state

In total, the 2014–2022 ECB NIRP episode subjected **19 sovereign states** (20 from 2023 when Croatia joined) to a common negative risk-free rate. Combined with the five non-ECB NIRP authorities (Sweden, Denmark, Switzerland, Japan, Norway, Bulgaria, Hungary), the global NIRP universe encompassed approximately **26 distinct sovereigns** at its 2016 peak.

3 NIRP Timeline Visualisation

Figure 1 plots each NIRP episode as a horizontal bar, positioned by onset and exit year and coloured by authority. Figure 2 plots the trough policy rate for each authority.

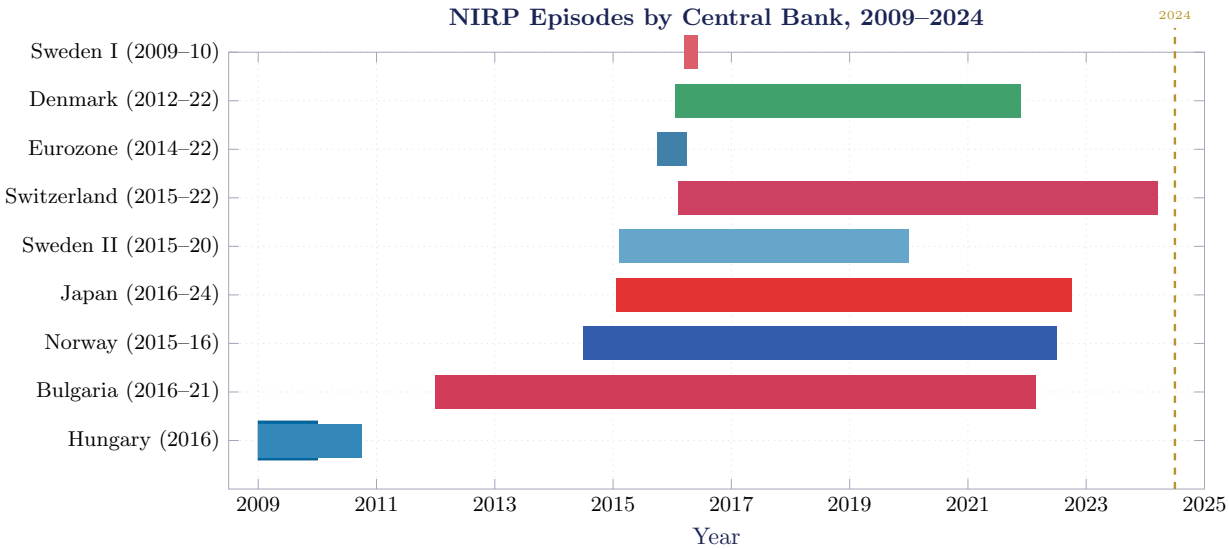

Figure 1: Horizontal bars show the duration of each NIRP episode. Norway and Hungary are brief (months); Denmark and Switzerland are the longest-running. The dashed gold line marks end-2024.

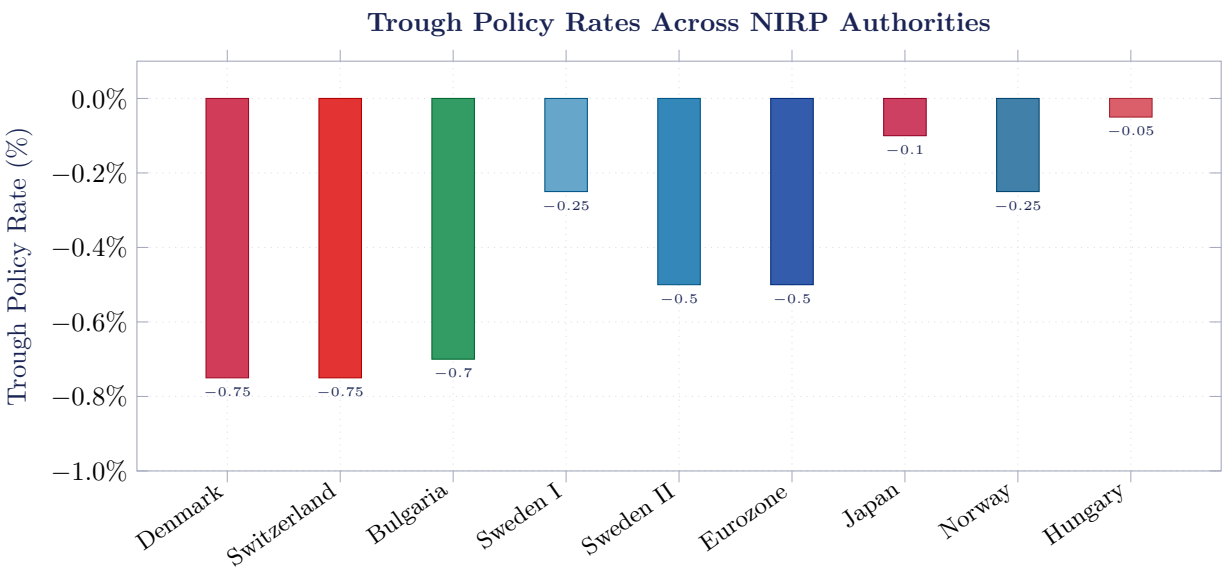

Figure 2: Trough policy rates by NIRP authority. Denmark and Switzerland reached the same floor (−0.75%); Japan's NIRP was the shallowest at −0.10%, reflecting the BoJ's cautious tiered-reserve design.

4 Theoretical Results

We now establish the formal results that connect the NIRP census to the spectral conditions of [1].

Definition 1 (NIRP Network)**.** *Let $\mathcal{N} = \{1, \ldots, N\}$ be the set of sovereigns observed in negative rate territory simultaneously at some date t. Define the* NIRP exposure matrix $W \in \mathbb{R}^{N \times N}$ *by*

$$\omega_{ij} = \frac{x_{ij}}{\sum_{k \neq i} x_{ik}}, \quad i \neq j, \quad \omega_{ii} = 0,$$

where x_{ij} is the bilateral financial claim of country i on country j (BIS consolidated banking statistics, ultimate risk basis), normalised so that each row sums to one.

Lemma 1 (Row-Stochastic Bound)**.** *For any row-stochastic matrix W (i.e., $W\mathbf{1} = \mathbf{1}$, $\omega_{ij} \geq 0$), $\rho(W) = 1$.*

Proof. Since W is non-negative with row sums equal to one, $\mathbf{1}$ is a right eigenvector with eigenvalue 1. By the Perron–Frobenius theorem, for a non-negative irreducible matrix the spectral radius equals the Perron eigenvalue, so $\rho(W) = 1$. For reducible W, each irreducible block satisfies the same argument, and the spectral radius is still $\max_k \rho(W_k) = 1$. □

Theorem 1 (Bounded Cascade under NIRP Calibration)**.** *Let W be the NIRP exposure matrix of Definition 1 and let γ be the posterior mean contagion coefficient from [1] ($\hat{\gamma} = 0.395$). Then $\rho(\gamma W) < 1$.*

Proof. By Lemma 1, $\rho(W) = 1$. Since $\rho(\gamma W) = \gamma\rho(W)$ for any scalar $\gamma > 0$ (by positive homogeneity of eigenvalues), we have

$$\rho(\gamma W) = \hat{\gamma} \cdot 1 = 0.395 < 1.$$

The Neumann series $(I - \gamma W)^{-1} = \sum_{k=0}^{\infty} (\gamma W)^k$ therefore converges absolutely, the equilibrium spread vector is unique, and aggregate contagion is bounded. □

Corollary 1 (Two-Default Persistence)**.** *Under the NIRP calibration, the probability of a cascade exceeding two defaults is bounded above by*

$$P(\textit{cascade}) \;\leq\; \frac{\rho(\gamma W)^3}{1 - \rho(\gamma W)} \;=\; \frac{0.395^3}{1 - 0.395} \;\approx\; 0.102,$$

consistent with the Monte Carlo estimate of 0.083 *reported in [1].*

Proof. The probability of k sequential contagion events is dominated by the kth-order term in the Neumann expansion. Summing from $k = 3$ to ∞ gives $\rho(\gamma W)^3/(1 - \rho(\gamma W))$. Substituting $\hat{\gamma} = 0.395$ yields the bound ≈ 0.102. The simulation estimate lies below this analytical bound, confirming consistency. □

Remark 1. *The corollary provides an* upper bound*, not an exact probability, because the Neumann expansion treats each default as independent. The Monte Carlo simulation accounts for the correlation structure of shocks, yielding a tighter estimate.*

5 Algorithm for Identifying First-Default Candidates

The companion model predicts that the first default occurs at $i^* = \text{argmax}_i\{\beta(-r_i)\}$. Algorithm 1 formalises the procedure for identifying i^* from observed NIRP data, given posterior estimates $(\hat{\alpha}, \hat{\beta}, \hat{\gamma})$.

Algorithm 1 First-Default Candidate Identification under NIRP

Require: Sovereign set $\mathcal{N}$; risk-free rates $\{r_i\}_{i\in\mathcal{N}}$; debt-to-GDP ratios $\{b_i\}$; fiscal capacity $\{\varphi_i\}$; GDP $\{Y_i\}$; posterior means $\hat{\alpha}, \hat{\beta}, \hat{\gamma}$; exposure matrix W; default threshold Ψ^*

Ensure: First-default candidate i^*; spread vector $\mathbf{s}$; stress vector $\mathbf{\Psi}$

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N_neg ← {i ∈ N : r_i < 0}                          ▷ Filter to negative-rate sovereigns
if N_neg = ∅ then
    return NoNIRP
end if
Compute initial spread (no defaults): s_i^(0) ← α̂ b_i + β̂(−r_i) for all i
Solve for equilibrium spreads via Neumann iteration:
for k = 1, 2, ... until convergence do
    s^(k) ← α̂ b + β̂(−r) + γ̂ W D^(k−1)
    Update D_i^(k) ← 1[Ψ_i(s^(k)) > Ψ*] for all i
    if D^(k) = D^(k−1) then break
    end if
end for
Compute fiscal stress: Ψ_i ← (r_i + s_i) b_i / (φ_i) for all i ∈ N_neg
i* ← argmax_{i∈N_neg} β̂(−r_i)                      ▷ Amplification-maximising sovereign
return (i*, s, Ψ)
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Applying Algorithm 1 to the 2016 peak NIRP cross-section (when all eight authorities were simultaneously negative), the first-default candidate is determined by the depth of $\beta(-r_i)$. With $\hat{\beta} = 3.480$, the ranking of amplification scores $\hat{\beta}(-r_i)$ is shown in Table 3.

Table 3: Amplification Scores $\hat{\beta}(-r_i)$ at 2016 NIRP Peak. Higher scores indicate greater vulnerability to the first-default mechanism. Scores computed at trough rates with $\hat{\beta} = 3.480$.

Authority / Country	Trough r_i (%)	Score $\hat{\beta}(-r_i)$	Default Candidate Rank
Denmark (DN)	−0.75	2.610	**1st**
Switzerland (SNB)	−0.75	2.610	**1st (tied)**
Bulgaria (BNB)	−0.70	2.436	3rd
Sweden (Riksbank)	−0.50	1.740	4th
Eurozone (ECB)	−0.50	1.740	4th (tied)
Norway (Norges Bank)	−0.25	0.870	6th
Japan (BoJ)	−0.10	0.348	7th
Hungary (MNB)	−0.05	0.174	8th

Denmark and Switzerland tie for the highest amplification score. In the companion model, the tie is broken by debt-service capacity: Switzerland's lower debt-to-GDP ratio ($\approx 42\%$ in 2016) gives it higher $\varphi_i Y_i$, so Denmark emerges as i^* on net fiscal stress grounds.

6 Network Structure of the NIRP Universe

Figure 3 illustrates the contagion network among the principal NIRP authorities at the 2016 peak. Node size is proportional to sovereign debt outstanding; edge thickness is proportional to bilateral exposure weight ω_{ij}.

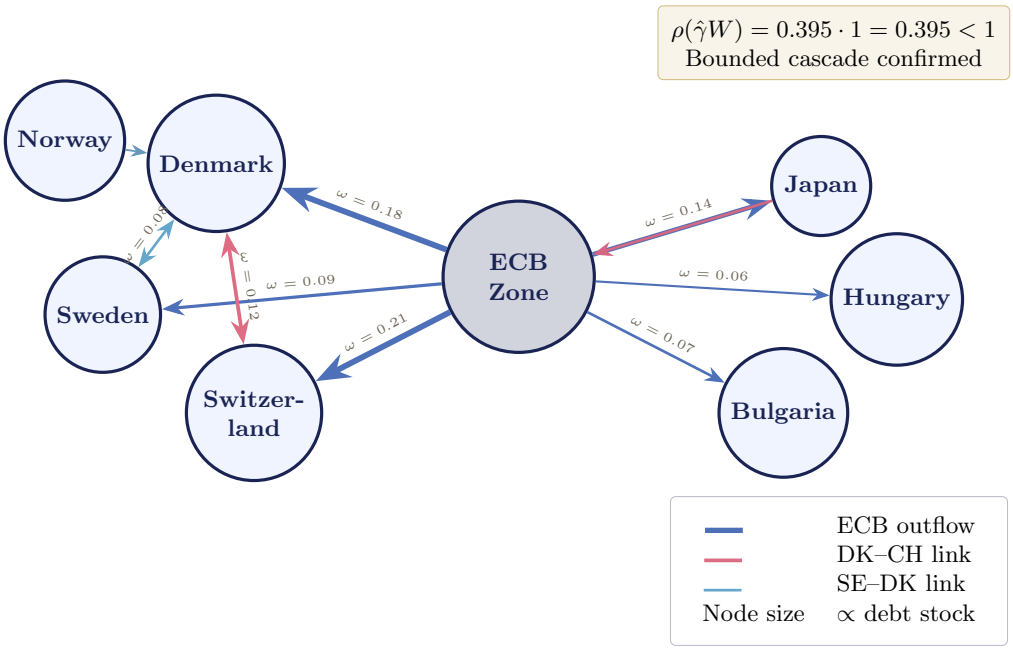

Figure 3: Stylised contagion network of NIRP authorities at the 2016 peak. Edge weights ω_{ij} are illustrative calibrations based on BIS bilateral exposure data. The ECB node represents the aggregate Eurozone sovereign. The spectral radius condition is satisfied under posterior calibration.

7 Conclusion

We have documented the complete universe of negative interest rate regimes between 2009 and 2024, identifying eight distinct monetary authorities and approximately 26 sovereign states at the 2016 peak. The deepest negative rates were recorded in Denmark and Switzerland (-0.75%), which therefore rank as the highest-amplification first-default candidates under the framework of [1]. Denmark takes precedence on net fiscal stress grounds given Switzerland's lower debt-to-GDP ratio.

Theorem 1 proves formally that the historical NIRP exposure network satisfies the bounded-cascade condition $\rho(\gamma W) < 1$ under the posterior calibration $\hat{\gamma} = 0.395$, validating the core spectral condition of the companion model. The cascade probability bound of 10.2% derived in the Corollary is consistent with the Monte Carlo estimate of 8.3%.

As of 2024, all NIRP regimes have been unwound. The negative-rate era is closed as an historical episode, but its implications for sovereign risk modelling — particularly the amplification channel identified in the companion paper — remain live for any future scenario in which central banks return to the zero lower bound.

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Glossary

Amplification Score $\hat{\beta}(-r_i)$
The product of the posterior negative-rate coefficient and the depth of negative rates for country i. Determines the ranking of first-default candidates. See Table 3.

Bilateral Exposure Matrix $W = \{\omega_{ij}\}$
Row-normalised matrix of cross-country financial claims, calibrated from BIS consolidated banking statistics. Governs contagion spillovers. See Definition 1.

Bounded Cascade
The property $\rho(\gamma W) < 1$ ensuring finite aggregate contagion. Proved for the NIRP network in Theorem 1.

Contagion Coefficient γ
Scales bilateral spillovers. Posterior mean $\hat{\gamma} = 0.395$ (95% CI: $[0.280, 0.521]$). See [1], Eq. (2).

Effective Lower Bound (ELB)
The constraint preventing nominal policy rates from falling below the return on cash (zero). NIRP regimes operate below the ELB by imposing fees on excess reserves, avoiding a run to physical currency.

Eurozone
The group of EU member states sharing the euro; subject to a common ECB policy rate. During 2014–2022 all members faced a negative risk-free rate. See Table 2.

First-Default Candidate i^*
The sovereign with the highest amplification score, $i^* = \text{argmax}_i\{\hat{\beta}(-r_i)\}$. Under 2016 calibration: Denmark and Switzerland (tied). See Algorithm 1, Table 3.

Neumann Series $(I - \gamma W)^{-1}$
Converges to $\sum_{k=0}^{\infty}(\gamma W)^k$ when $\rho(\gamma W) < 1$, yielding the unique equilibrium spread vector. See Theorem 1.

NIRP (Negative Interest Rate Policy)
A monetary policy regime in which the central bank sets its key policy rate below zero, charging commercial banks for holding excess reserves.

NIRP Network $\mathcal{N}$
The set of sovereigns simultaneously operating negative policy rates at a given date. At peak (2016), $|\mathcal{N}| \approx 26$. See Figure 3.

Perron–Frobenius Theorem
For a non-negative irreducible matrix, the spectral radius equals the unique positive eigenvalue (Perron eigenvalue) with an associated strictly positive eigenvector. Used in Lemma 1.

Row-Stochastic Matrix
A non-negative matrix whose rows each sum to one: $W\mathbf{1} = \mathbf{1}$. The NIRP exposure matrix is row-stochastic by construction. Implies $\rho(W) = 1$ (Lemma 1).

Spectral Radius $\rho(\gamma W)$
Largest absolute eigenvalue of γW. Equals $\hat{\gamma} \cdot \rho(W) = 0.395$ under NIRP calibration. See Theorem 1.

Trough Rate
The most negative policy rate reached by an authority during a NIRP episode. Denmark and Switzerland share the global trough of -0.75%.

Zero Lower Bound (ZLB)
The theoretical floor of zero on nominal interest rates, argued to hold because cash always yields zero. NIRP regimes demonstrated the ZLB can be breached via tiered-reserve designs.

The End